Assignment 3

Task 1

1. The key design challenges in the research paper are scalability, robustness, extensibility, manageability, portability and a low overhead.
2. It is based on a hierarchical design that relies on a listen/announce protocol to monitor the state within clusters. Within each cluster every node runs gmond which is the ganglia monitoring protocol. The gmonds have threads for dividing the work being done inside of them. Then the clusters use gmetad which is the ganglia meta daemon, to provide federation of multiple clusters. Then gmetric, which is a command-line program, is used to publish application-specific metrics.
3. One improvement is suggested in the paper and that is that when a new node joins or restarts the program could let new members bootstrap directly from the eldest gmond in the multicast group instead of relying purely on multicast-based rebuilding of state.

Task 2

1. There were three drivers for self-adaptation: Adapting to changes in higher-level goals, adapting to changes in the system and adapting to changes in the environment. Higher-level goals mean that the system may adapt due to changes in the objectives. Changes in the system means that it adapts to changes in component behavior or other couplings between subsystems. Changes in the environment means that it can adapt due to things like safety-critical situations or weird interactions.
2. Control theory is a technique that is based on the feedback of the system. The controller measures what state the system is in and adapts according to a wanted ideal state.
   Machine learning is a technique for teaching the system from experience. It differs from control theory by instead of just aiming for an ideal state trying to learn from changes in outcomes due to changes in parameters.
   Optimization and operations research is a technique that is based on calculating the best configuration for the system with the model, constraints and parameters.
3. Control theory can be used in a data center by regulating the computing-system behavior and by controlling the temperature. The temperature is an important constraint in data centers and by aiming for an ideal state the control theory can make sure that it maximizes performance while staying cool.
   Machine learning can be used to balance the performance and the power consumption of the data center. It can compare the performance at different power consumption levels and decide what is the best configuration for the wanted level of power consumption.
   Optimization and operations research can be used for parts of the data center such as the load and data storage allocation by calculating the best possible configuration from the existing constraints and parameters.